Environ Model Assess
DOI 10.1007/s10666-014-9409-3

Modeling a Carbon Capture, Transport, and Storage Infrastructure for Europe

Pao-Yu Oei · Johannes Herold · Roman Mendelevitch

Received: 14 March 2012 / Accepted: 31 March 2014
© Springer International Publishing Switzerland 2014

Abstract In this paper, we develop a model to analyze the economics of carbon capture, transport, and storage (CCTS) in the wake of expected rising CO_2 prices. We present a scalable mixed integer, multiperiod, welfare-optimizing network model for Europe, called CCTS-Mod. The model incorporates endogenous decisions on carbon capture, pipeline and storage investments, as well as capture, flow and injection quantities based on given costs, CO_2 prices, storage capacities, and point source emissions. Given full information about future costs of CCTS-technology, and CO_2 prices, the model determines a cost minimizing strategy on whether to purchase CO_2 certificates, or to abate the CO_2 through investments into a CCTS-chain on a site by site basis. We apply the model to analyze different scenarios for the deployment of CCTS in Europe, e.g., under high and low CO_2 prices, respectively. We find that beyond CO_2 prices of €50 per t, CCTS can contribute to the decarbonization of Europe's industry sectors, as long as one assumes sufficient storage capacities (onshore and/or offshore). We find that CCTS is only viable for the power sector if the CO_2 certificate price exceeds €75 per t.

Keywords Carbon capture · CCS · Modeling · Infrastructure · CO_2-mitigation · MIP

P.-Y. Oei (✉) · J. Herold · R. Mendelevitch
TU Berlin, Workgroup for Economic and Infrastructure Policy, Straße des 17. Juni 135, 10623, Berlin, Germany
e-mail: pyo@wip.tu-berlin.de

J. Herold
e-mail: johannes.herold.privat@gmail.com

R. Mendelevitch
e-mail: rm@wip.tu-berlin.de

1 Introduction

The ongoing carbon capture, transport, and storage (CCTS) discussion originates from multiple perspectives: on the one hand, longer-term energy system models insist on the need of CCTS to achieve ambitious decarbonization scenarios (e.g., [22]). On the other hand, progress in advancing the technology on the ground has been modest thus far, particularly in Europe and the US [18, 19]. The IEA underlines in its "Energy Technology Perspectives 2012" study that its importance with an overall 20 % contribution to achieving emission reduction goals and a 40 % cost increase in the absence of the technology [23]. At the same time, they acknowledge the real danger that the ambitious development plans for CCTS demonstration in Europe will remain unfulfilled. Among other concerns, the institutional question about regulatory and environmental issues with storage could substantially hinder the deployment. In December 2012, the European Commission decided not to consider any CCTS project in the first round of the NER300 funding program, but supporting 23 renewable energy projects with 1.2 bn., instead. The lack of financial guarantees from project partners and member states as well as insufficiently advanced project status highlighted the uncertain future of CCTS in Europe [8].

To date, the discussion has centered on the role of CCTS in the power sector [46], yet the technology also holds promise for the iron and steel, cement, as well as refining sectors where chemical processes emit large amounts of CO_2. Switching to renewable sources and/or increasing process efficiency will result in partial emissions reductions in the medium term, e.g., 35 % in the iron and steel sector, 35 % in cement, and 20 % in clinker production [31]. Low-carbon substitutes to the conventional production of these raw materials, such as magnesium cement

Published online: 30 May 2014

or the electrolytic production of iron, may become available in the future. However, the extent to which they could be applied on a large scale as well as prove economically viable is unknown. Thus the CCTS technology remains the only short- and mid-term CO_2 mitigation option for these sectors. At the same time, an application in these sectors will lead to lower capture costs than in the energy sector due to the higher CO_2 concentration in the flue gas [20, 31]. Despite this fact, industrial partners have made only little effort to bring forward CCTS projects. Most industrial companies also lack the financial possibilities to invest into a demonstration unit including transport and storage of CO_2. One major argument against putting pressure on industrial facilities in Europe is the fear of losing international competitiveness when facing higher production costs due to CCTS. This apprehension was, for example, present in the design of the allocation scheme for EU-ETS emission allowances. The pure grandfathering approach did not put any pressure on the emission efficiency of existing facilities (as widely criticized, e.g., in [26]) and thus free allowances were used instead of pushing for CCTS. The only two large-scale industry CCTS demonstration projects in Europe, ULCOS Florange (a steel making plant in Lorraine, France), and Green Hydrogen (a hydrogen plant in Rotterdam, Netherlands) initially applied for NER300 funding, but then, in 2012, withdrew their application [39]. It is worth noting that those industry CCTS projects that are currently operating face favorable and very site-specific conditions. Either CO_2 capture is disproportionally inexpensive due the specific process (e.g., ethanol production, in Decatur, Illinous, USA), or the CO_2 has to be captured regardless in order to market the product (e.g., natural gas with a too high CO_2-concentration as in, e.g., Sleipner field, Norway), or additional revenue from CO_2-enhanced oil recovery changes the economics of the CCTS project (e.g., Weyburn Project in Saskatchewan, Canada). Hirschhausen et al. [19] analyzed the discrepancy between the hopes put into the technology and its current state of development. In Addition to the points mentioned above, they found that there was a lack of technological focus on cheap capturing technologies. Also, blue sky expectations on cost reductions and learning curves, as well as the fact that the costs and complexity related to regulatory issues of CO_2 transport and regulatory and technological issues of CO_2 storage were neglected. Moreover, persisting negotiations and complicated environmental assessments for CO_2 storage fueled by "not in my backyard" (NIMBY) concerns hindered the implementation of planned demonstration projects.

Against this background, the question arises of what contribution the CCTS technology can realistically make toward European CO_2 emission reduction.

We apply the CCTS-Mod model to analyze the potential development of a CCTS infrastructure in Europe. In particular, we investigate in the nature of the CO_2 transport infrastructure that would emerge in Northwest Europe, i.e., in Germany and to its south and east, ranging to France and up to the North Sea and its neighboring states. Several scenarios, differing by the estimate of geological storage available, the availability of onshore storage, and the expected CO_2 certificate price in 2050, are run. We find that under certain extreme assumptions, such as a relatively high CO_2 price, and very optimistic CO_2 storage availability, a large-scale CCTS roll-out might indeed be expected. However, in a more realistic scenario, including lower storage availability and public resistance to onshore storage, a large-scale roll-out is not going to happen. In all scenarios, CCTS deployment is highest in CO_2 intensive non-energy industries, where emissions cannot be avoided by fuel switching or alternative production processes.

The next section provides an overview of existing literature and models, both theoretical and applied, e.g., to North America or Europe. Section 3 specifies our own model, called CCTS-Mod and its data. We then apply CCTS-Mod to analyze the potential development of a CCTS infrastructure in Europe under certain scenarios. Section 5 summarizes the findings and provides conclusions.

2 Model

Recent literature points out that the real bottlenecks to CCTS deployment are transport and storage infrastructure [17]. Against this background, only a few simplified CCTS models actually address the pipeline transport of large volumes of CO_2. The Global Energy Technology Strategy Program (GTSP) modeled the adoption of a CCTS system within three fossil-fuel-intensive electricity generation regions of the US. The results show that CCTS implementation depends more on CO_2 injection rates and total reservoir capacity than on the number of potential consumers who would use the CO_2 for enhanced oil recovery (EOR) [7].

McPherson et al. [35] and Kobos et al. [32] introduced the "String of Pearls" concept to evaluate and demonstrate the means for achieving an 18 % reduction in carbon intensity by 2012 in Texas using CCTS. Their dynamic simulation model connects each CO_2 source to the nearest sink and automatically routes pipelines to the next neighboring sink, thus creating a trunkline connection for all of the sinks. While the model can determine an optimal straight-line pipeline network, it is not possible to group flows from several sources to one sink. Fritze [11] developed a least-cost path model connecting each source with the nearest existing CO_2 sink. The paper examines a hypothetical case of main trunk lines constructed by the US Federal Government

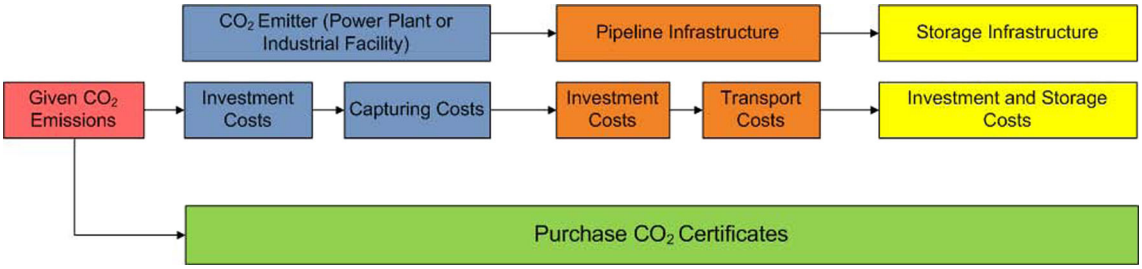


Fig. 1 Decision tree in the CO_2 disposal chain of the CCTS-Mod

and its influence on the total costs. However, no economies of scale are implemented for construction, as the costs of building the public trunk lines are greater than the potential costs of private enterprises. Nevertheless, public trunk lines allow greater network flexibility and redundancy which can lead to cost savings in times of emergency and when storage capacity needs to be balanced.

Middleton et al. [37] designed the first version of the scalable infrastructure model SimCCS based on mixed integer linear programming (MILP). With its coupled geospatial engineering-economic optimization modeling approach, SimCCS minimizes the costs of a CCTS network capturing a given amount of CO_2. An updated version by Middleton and Bielicki [38], comprising of 37 CO_2 sources and 14 storage reservoirs in California, simultaneously optimizes the model according to the amount of CO_2 to be captured from each source; the siting and construction of pipelines by size; and the amount of CO_2 to be stored in each sink. The decisions are endogenous, but the total amount of CO_2 to be stored is exogenous. Economies of scale are implemented via possible pipeline diameters in 4-in steps, each with its own cost function. Kuby et al. [33] extend a smaller version of the model that employs 12 sources and 5 sinks in California with a market price of CO_2 as well as a benefit when used in EOR. This model minimizes the costs of CCTS, but only examines one period. Their findings of a CO_2 price sensitivity analysis indicate that infrastructure deployment is not always sensitive to the price of CO_2. Kazmierczak et al. [28] and Neele et al. [40] develop an algorithm to create a low-cost network and a decision support system to evaluate the economical and technical feasibility of storage. A realistic estimate of the economic feasibility of a potential CCTS project is possible, but there is no detailed planning at the project level.

In summary, only a few models include economies of scale in the form of possible trunk lines, but they operate on a static level or are based on an exogenously set amount of CO_2 to be stored. Therefore, the models exclude the option of buying CO_2 certificates instead of investing in CCTS infrastructure. We introduce a scalable mixed integer, multi-period, welfare-optimizing CCTS network model, hereafter CCTS-Mod. The model incorporates endogenous decisions on carbon capture, pipeline and storage investments, as well as capture, flow, and injection quantities based on exogenous costs, a CO_2 certificate price path, a comprehensive set of emissions point sources from European power and industry sectors, as well as on- and offshore storage sites in depleted hydrocarbon fields and saline aquifers. Our model runs in 5-year periods beginning in 2005 and ending in 2050. Capacity extensions can be used in the period after construction for all types of investments in the model. Sources and sinks are linked to nodes according to their geographical position and pipelines are constructed between neighboring nodes. To ensure a better resolution, no aggregation of sources/sinks at a node takes place. The distance between two neighboring nodes can be chosen flexibly, making CCTS-Mod scalable and thus allowing different degrees of resolution. Economies of scale are implemented by discrete pipeline diameters with respective capacities and costs.

2.1 CCTS Decision Tree

Figure 1 illustrates the decision path of CCTS-Mod based on the CO_2 disposal chain when using the CCTS technology. Each producer of CO_2 must decide whether to release it into the atmosphere or store it via CCTS. The decision is based on the price for CO_2 certificates and the investment required for the capture unit, the pipeline and the storage facilities, and the variable costs of using the CCTS infrastructure.

We apply a stylized institutional setting to a vertically integrated CCTS chain. A single omniscient and rational decision-maker has perfect foresight and makes all investment and operational decisions.[1] Under these simplifying assumptions, we run the model using a single cost minimization.

[1] The model tends to overestimate the potential for CCTS. Considering the large number of different players and technologies, the uncertainties regarding CO_2 prices, learning rates, legal issues, permitting, certification of storage capacity, and further policy measures would increase the total costs. Real costs are therefore expected to be higher and come along with a lower deployment of CCTS in the future.

2.2 Mathematical Formulation

We define the objective function 1 to be minimized as follows:

$$\min_{\substack{x_{Pa},inv_x_{Pa},\\ z_{Pa},f_{jia},inv_f_{ijda},\\ plan_{ija},y_{Sa},inv_y_{Sa}}} \sum_a \left\{ \left(\frac{1}{1+r}\right)^{(year_a - year_{start})} \cdot \left\langle \sum_P \left[(c_ccs_{Pa} + (1 - capt_rate) \cdot cert_a) \cdot x_{Pa} + c_inv_x_P \cdot inv_x_{Pa} + cert_a \cdot z_{Pa} \right] + \sum_i \sum_j \left[E_{ij} \cdot \left(c_f \cdot f_{ija} + \sum_d \left(c_inv_f_d \cdot inv_f_{ijad} \right) + c_plan \cdot plan_{ija} \right) \right] + \sum_S \left[c_inv_y_{Sa} \cdot inv_y_{Sa} \right] \right\rangle \right\} \tag{1}$$

with:

$$x_{Pa}, inv_x_{Pa}, z_{Pa}, f_{jia}, y_{Sa}, inv_y_{Sa} \geq 0 \tag{2}$$

$$inv_f_{ijda} \in N_0 \tag{3}$$

$$plan_{ija} \in [0; 1] \tag{4}$$

The first term of the objective function 1 is the discount factor, where r is the interest rate, $year_a$ is the starting year of period a, and $start$ is the starting year of the model. From here, we can split the objective function 1 into three parts representing the three steps of the CCTS chain. For the first step, the decision variables are the dimensioning of the capture system inv_x_{Pa} and the current level of CO_2 emission that are cycled through the capturing system ($x_{Pa} \cdot capt_rate$ represents the amount of CO_2 actually captured by the facility). An individual variable is declared for each emitter P in each period a. The parameter $capt_rate$ represents the maximal possible percentage of captured CO_2, thus certificates still have to be purchased at the price of $cert_a$ for the remaining fraction.

The second part represents the transportation step. The decision variables are as follows: f_{ija} declares the CO_2 flow from node i to j in period a; inv_f_{ijda} denotes the number of pipelines to be built between node i and j with the diameter d in period a; $plan_{ija}$ is a binary variable (see Eq. 4) and has the value one if a pipeline route between node i and j is planned and licensed in period a, and zero otherwise. Routing of pipelines is a central aspect of our study, and we implement a detailed process of pipeline building by introducing the planning variable and, thus, separate the planning and development costs from the rest of the capital costs. Additional pipelines on already licensed routes do not face licensing or planning costs. The desired effect is that new pipelines are routed along existing lines as observed in reality.

The third part represents storage. The decision variables are y_{Sa}, which is the quantity stored in storage facility S in period a, and inv_y_{Sa}, which denotes the investments in additional annual injection capacity. Variable costs of CO_2 storage are considered negligible as they sum up to less than 7–8 % of the overall storage costs (see Section 3.3 for further explanations).

All decision variables have to be non-negative (see Eq. 2). Additionally, the number of pipelines to be build on one route inv_f_{ijda} are discretized (see Eq. 3).

In the objective function 1, each decision variable is multiplied by its respective cost factor. E_{ij} is a distance matrix indicating whether two nodes i and j can be connected directly. If possible, the values of the matrix give the distances between i and j in kilometers. Scaling is easily done by varying the distance between nodes and their number and the spatial focus can range from regional to worldwide depending on research question and existing data sources.

The model is restricted by:

$$x_{Pa} + z_{Pa} = CO2_{Pa} \qquad \forall P, a \tag{5}$$

Equation 5 defines that a facility's CO_2 stream can be treated in two ways, or a mixture of it: CO_2 emissions can either be balanced with CO_2 certificates (z_{Pa}), or the CO_2 can be cycled through a capture system (x_{Pa}). Note that even if the entire CO_2 stream is treated in the capturing facility (i.e., x=CO_2) a fraction of (1-$capt_rate$) $\cdot$ x_{Pa} is released into the atmosphere and needs to be balanced with CO_2 certificates (c.f. Eq. 1).

$$\sum_i f_{ija} - \sum_i f_{jia} + \sum_P \left(match_P_{Pj} \cdot x_{Pa} \cdot capt_rate \right) - \sum_S \left(match_S_{Sj} \cdot y_{Sa} \right) = 0 \qquad \forall j, a \tag{6}$$

Equation 6 specifies the physical balance condition, which states that all flows feeding into a node j must be discharged from the same node. $match_P_{Pj}$ declares whether producer P is located at node P_j, while $match_S_{Sj}$ declares whether a sink S is located at node S_j. The amout of CO_2 that is transported and stored through the system is equal to the amout acctually captured at the respective facility ($x_{Pa} \cdot capt_rate$).

$$x_{Pa} \leq \sum_{b<a} (inv_x_{Pb}) \qquad \forall P, a \tag{7}$$

The capturing capacity of each producer P in period a is given in Eq. 7. Note that all terms in this inequality are decision variables, meaning that injection in period a can only happen if the capacity was expanded prior to period a.

$$f_{jia} \leq \sum_{b<a} \sum_{d} \left(cap_d_d \cdot inv_f_{ijdb}\right) + \sum_{b<a} \sum_{d} \left(cap_d_d \cdot inv_f_{jidb}\right) \qquad \forall i, j, a \qquad (8)$$

The capacity restriction of the pipeline in Eq. 8 works similarly to Eq. 7.

$$y_{Sa} \leq \sum_{b<a} inv_y_{Sb} \qquad \forall S, a \qquad (9)$$

Inequality 9 states that the annual injection rate of a storage facility S is limited to the sum of investments in injection capacity inv_y_{Sb} from previous periods b.

$$\sum_{a} (5 \cdot y_{Sa}) \leq cap_stor_S \qquad \forall S \qquad (10)$$

Inequality 10 restricts the amount of CO_2 injected into reservoir S to its physical capacity. The multiplication by 5 resembles the amount of years per period a.

$$\sum_{d} \left(inv_f_{ijda}\right) \leq max_pipe \cdot \sum_{b<a} \left(plan_{ijb}\right) \qquad \forall i, j, a \qquad (11)$$

Planning, licensing, and optimal routing of pipelines is ensured via Eq. 11 where max_pipe is the maximum number of pipelines that can be built on a licensed route.

The model is solved in the general algebraic modeling system (GAMS) using the CPLEX solver.

3 Data

3.1 CO_2 Emission Sources

Our European emission data covers the EU27 plus Switzerland and Norway. It includes industry facilities from iron and steel production, the cement and clinker production as well as oil refineries. Furthermore, waste-, natural gas-, lignite-, and coal-fueled power plants that emit more than 100,000 t CO_2 per year are included. Facilities below this emission level are considered too small to justify the investment into capture, transport, and storage. Data on the average annual CO_2 emissions of individual plants, location, and age are taken from Platts [42] and EEA [9]. We assume a lifetime of 55 years for lignite and hard-coal plants and 40 years for NGCC plants. Industrial facilities are assumed to be reconstructed with the same characteristics and on the same site once their economical lifetime ends. Projections on new power plant capacity installation are taken from VGB [48], covering 66 GW of new natural gas (NGCC), 7.6 GW of lignite, and 35 GW of coal plants. Due to capacity aging and scrapping of old plants, this results in a decrease in fossil fuel capacity until 2050.

The total number of implemented emission sources in 2010 totals 2,725 facilities, with emissions of 2.122 $GtCO_2$ annually. These divide into 1,476 (1.527 $GtCO_2$/a) fossil-fueled power plants and 1,249 (0.595 $GtCO_2$/a) industrial facilities. The graphical distribution of the included point sources is shown in Fig. 2.

The CCTS investment costs for the sectors considered in this paper are presented in Table 1. Costs estimates for the power generation sector are available from various sources [10, 27, 47, 49, 50]. They all share the same general trend of slower capital cost for coal-fired generation compared to gas-fired power plants when calculated in € per tCO_2. In the more recent studies, a great share of the variation in the cost figures is attributed to changing raw material prices and different assumptions on the risk premium attributed to this immature technology. Costs for industry capture are gaining increased attention (see, e.g., [31, 34]). Rubin et al. [45] examines learning rates of different climate protection technologies and estimates learning rates for carbon capture that we apply to our data from 2020 onwards.

The variable costs of capture have two components: variable costs of operation and maintenance and an energy penalty for additional energy input needed for the capturing process. Total costs are given in Table 2. For coal and lignite plants, we apply the post-combustion capture technology. We assume the oxy-fuel process for the iron and steel and the cement sectors as proposed by Öko Institut [31]. In this case, the variable costs of capture are mainly driven by the price for the electricity needed for the air separation unit. We assume a fixed price of €70/MWh, which remains constant. In refineries, we assume post-combustion capture. Due to the low CO_2 concentration in the flue gas and the high diversity of the fuels and processes used in refineries, variable costs are comparable to those in natural gas plants [20].

3.2 CO_2 Transport

Pipeline transportation is commonly considered as the only economically viable onshore transport solution that can carry the quantities emitted by large-scale CO_2 sources. Onshore transport faces few technological barriers due to experience in the gas and oil sector and the CO_2 industry for enhanced oil recovery (EOR) in the US. CO_2 pipelines represent a typical network industry and are characterized by high upfront, sunk investment costs. Variable costs are comparatively insignificant and primarily include expenditures for fuelling compressors.

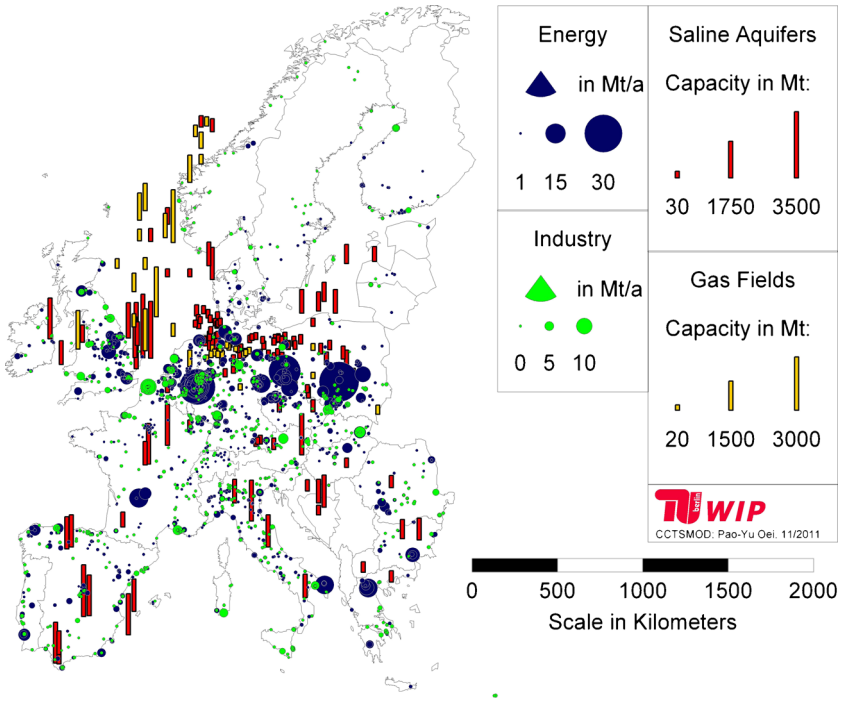


Fig. 2 Emission sources and storage potential

According to Heddle et al. [16], right of way (ROW) costs account for 4 to 9 % of total gas pipeline construction costs depending on the diameter of the pipe, which we used to derive our values of the plan parameter. ZEP [51] presents a comprehensive study on CO_2 transportation costs for different setting of transport networks. Calculated transport costs in € per tCO_2 range from 2 to 20 depending on the network setting. Associated capital costs range between €0.08 and 0.15 per tCO_2 and kilometer of pipeline. Topographic features, such as mountains or densely populated areas, are often neglected in studies as they need additional data and increase the computing time. Including such features, however, would lead to a strong increase of the transport costs or even infeasibilities of some projects (e.g., due to mountain ranges).

To account for the uncertainty associated with topographic features, public resistance, and environmental concerns as uncertain utilization rates, we employ a higher value of €0.087 per tCO_2, cm of pipeline diameter, and km of pipeline. Economies of scale associated with CO_2 pipeline transport pipelines are depicted through the five possible diameters with associated annual transport capacity (see Table 3).

For O&M costs, ZEP [51] give values of €0.005 to 0.01 per tCO_2 and kilometer. IEA [21] arrive at similar operation costs varying between €0.01 and 0.025 per km and year depending on pipeline diameter and total pipeline length, including costs for booster stations; we thus use a value of €0.01 per year and km and t CO_2 transported. Including the flow-dependent cost

Table 1 Investment costs for capture facilities in € per annual tCO_2 emissions (dimensioning of capturing system)

Technology	2010	2020	2030	2040	2050
Coal	150	150	139	119	93
Lignite	116	116	107	92	72
NGCC	275	275	255	218	171
Cement	135	135	125	107	84
Iron and Steel	117	117	108	93	73
Refineries	210	210	195	167	131

Source: Own calculation based on Tzimas [47], Ho et al. [20], Öko Institut [31], and Rubin et al. [45].

Typically, investment and operating costs for CO_2 capture are given in terms of MW and MWh, respectively. These figures refer to a specific capture rate (i.e., when making this investment one is able to capture a portion of the CO_2 otherwise emitted into the atmosphere). The basic unit of the CCTS-Mod is tCO_2. Thus, we converted the figures accordingly to arrive at a per tCO_2 based figure.

Table 2 Variable costs in € per tCO_2 treated in the capturing system

Technology	2010	2020	2030	2040	2050
Coal	32	32	31	31	31
Lignite	29	29	29	29	28
NGCC	47	47	45	44	44
Cement	17	17	17	17	17
Iron and Steel	16	16	16	16	16
Refineries	47	47	45	44	44

Source: Own calculation based on Tzimas [47], Ho et al. [20], Öko Institut [31], and Rubin et al. [45].

component ensures the shortest possible routing for the CO_2.

Planning and development (P&D) costs include ROW costs, land purchase, and routing costs which occur only for the first pipeline built on a certain route. This leads to the construction of pipelines along corridors.

3.3 CO_2 Storage

Data on CO_2 storage is difficult to come by and verify. Using available data, we derived our own estimates of location and capacity of the European on- and offshore storage. The exact location of the storage fields is being modeled as closely to the geological formation as possible. Bentham [3], Bentham et al. [4], Brook et al. [5], and Hazeldine [15] are used to get data for the UK and for offshore storage beneath the North Sea. Greenpeace [14] give good estimates for storage potential in Germany, while Radoslaw et al. [43] focusses on Poland. The feasibility study for Europe-wide CO_2 Infrastructure from the European Commission [1] and the Geo Capacity project [12] are used to estimate storage potential when no more accurate country-specific study was available to the public. These studies, however, only grant public access to storage data on a 50 x 50 km grid. This means that some of these formations might consist of several smaller neighboring aquifers. The example of Germany shows that the majority of the aggregated storage potential can actually be found in small reservoir of 50 Mt or less [14]. The exploration of such very small reservoirs is uneconomical, given a bad ratio of investment costs and exploitable storage capacity. The overall storage potential of Europe is thus overestimated due to the lack of more detailed information. The total storage capacity is set to 94 Gt, spread among 41 Gt saline aquifers and 3 Gt depleted gas fields onshore and, offshore, 30 Gt saline aquifers and 20 Gt depleted gas fields (see Table 8 in Appendix).

Table 3 Investment cost by pipeline diameter and respective annual transport capacity

Diameter (m)	Annual transport capacity (mio tCO_2 / a)	Investment costs (€ per tCO_2 and km)
0.2	6	0.29
0.4	18	0.19
0.8	71	0.10
1.2	174	0.06
1.6	338	0.04

Source: Own calculations based on IEA [21] and Ainger et al. [1]

According to Heddle et al. [16], the costs for CO_2 storage are determined by factors including type of storage facility, storage depth, permeability, number of injection points, injection pressure, etc. Therefore, total storage costs vary significantly in different studies [44]. A characteristic value for a storage project is the sum of costs per injection well including site development, drilling, surface facilities, and monitoring investments for a given annual CO_2 injection rate. Storage investments exhibit a strong sunk cost character and according to IEA [21] variable costs total only 7 to 8 %. Therefore, we implement storage costs on a total costs basis (see Table 4). A more recent estimate of storage costs from IEAGHG and ZEP [25] examining different settings and uncertainties on technological and regulatory issues arrive at figures similar to those presented above.

One option that is said to improve the economics of CO_2 storage and CCTS in general is CO_2-enhanced oil recovery [24]. The technology is increasingly used in the US and Canada [39] and might also be an option to provide additional investment incentive for CCTS projects in Europe. Studies look intensively into the interaction of these two technologies. Some regional studies on the UK and Norwegian potential [29, 30] as well as larger scope studies (e.g., on the North Sea region [36] or the US [6]) are also available. In general, it is up to future research to determine whether the combination of the two technologies can still be considered as CO_2 abatement, when taking into account the emissions from additionally recovered oil and assessing different injection strategies (see, e.g., [2]). For our approach, we do not consider CO_2-EOR as a storage option.

4 Scenarios and Results

The level of uncertainty about the size and configuration of the pipeline network emanates from the uncertainty about

Table 4 Site development, drilling, surface facilities, and monitoring investment cost for a given annual CO_2 injection rate per well

Technology	Natural gas field		Saline aquifer	
	Onshore	Offshore	Onshore	Offshore
Drilling length (m)	3,000	4,000	3,000	4,000
Well injection rate ($MtCO_2$/a)	1.25	1.25	1	1
Corrected well injection rate ($MtCO_2$/a)[a]	0.4	0.4	0.33	0.33
Drilling costs (€per m)	1,750	2,500	1,750	2,500
Investment in surface facilities (M€)	0.4	25	0.4	25
Monitoring investments (M€)	0.2	0.2	0.2	0.2
Wells per location	6	6	6	6
Total drilling costs (M€)	5.25	10	5.25	10
Total capital costs per well (M€)	5.6	14.5	5.6	14.5
O&M and monitoring costs (%)	7	8	7	8

[a]According to [13], an annual injection rate of 300,000 to 400,000 tCO_2 per well is more realistic for most formations. Source: Own calculations based on IEA [21]

future carbon policies, the level of deployment of renewable energy technologies, as well as the suitability and usability of geological formations to store captured CO_2. Different scenarios are implemented with a linear increase in CO_2 prices until 2050. For the base case, the CO_2 certificate price increases from €15 per t in 2010 to €75 per t in 2050. Additionally, we define a scenario with a higher (€100 per t) and a lower (€50 per t) CO_2 price in 2050. We do not implement a correlation between CCTS deployment and the price for CO_2, as it imposes a political variable. We also consider the possibility that onshore storage may not be possible in Europe, due to technical, political, or whatever other reasons. In that case, storage would need to take place offshore, mainly in the North Sea, and the total storage potential would be significantly reduced, from 94 Gt (on- and offshore) to only 50 Gt.

Table 5 Key scenario assumptions

Scenario	CO_2 price in 2050 (€/tCO_2)	Storage on/ offshore	CO_2 storage capacity (Gt)
Ref75	75	On and offshore	94
Off75	75	Offshore	50
On50	50	On and offshore	94
Off50	50	Offshore	50
On100	100	On and offshore	94
Off100	100	Offshore	50

The respective scenario key assumptions are shown in Table 5.

4.1 Reference Scenario: Certificate Price Increasing to 75 €/tCO2 in 2050

4.1.1 On- and Offshore Storage

Our reference scenario simulates the cost-optimal deployment of a European CCTS infrastructure for the period 2010–2050 given a CO_2 certificate price starting at €15 in 2010 and rising to €75 in 2050. Point source emissions, storage sites and potential pipelines are mapped on a spherical grid covering Europe. The distance between two neighboring grid nodes is 2° (on average about 200 km).

In this reference scenario, 758 Gt of emissions are captured, transported, and stored through CCTS in 2050. CCTS implementation begins in 2020 with the first investments before the capturing process starts 5 years later in both the iron and steel as well as in the cement sectors. CCTS infrastructure gradually ramps up from 2020 to 2040 (see Fig. 3). At first, the industrial facilities with low capturing costs situated close to potential storage sites are the predominant users of CCTS. Industrial CCTS penetration reaches saturation with a capturing rate of 370 Mt CO_2 per year in 2050. With rising CO_2 prices, CCTS becomes a more attractive abatement option for the power sector. The annual rate of stored CO_2 from power generation reaches 390 Mt in 2050 (see Table 6).

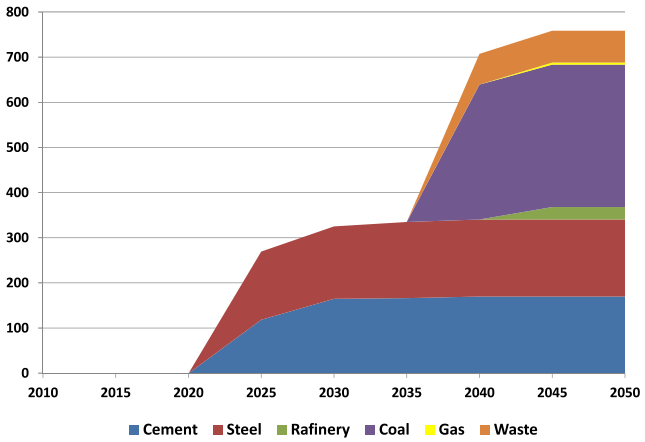


Fig. 3 Storage by sectors in million tCO_2, Ref75

Over the 40 year modeling time horizon, total investment costs along the CCTS value chain sup up to 240 billion euros. Given a total quantity of avoided emissions of 15.8 $GtCO_2$, this breaks down to investment cost of €15.3 per tCO_2 avoided. Total variable costs sum up to 515 billion euros, or €33 per tCO_2 avoided. Although this number may appear low, we note that most capture occurs in the industrial sector in the early years. The costs of the capturing process hereby comprise around 90 % of the total costs while the transport and storage only have minor impacts assuming an optimal grid and storage planning (see Fig. 4).

We note that under the applied CO_2 price path, CCTS is an option primarily for countries with a regional proximity between CO_2 intensive regions and storage sites. The technology is mostly implemented by Poland, Germany, the Netherlands, Belgium, France, and the UK. Moreover, we find no interconnected, transnational transportation network (see Fig. 5). As industry facilities will be the first-movers, they drive the layout of the pipeline network.

4.1.2 Offshore Storage Only

Due to longer transport distances and more expensive storage, this subscenario leads to a deployment of CCTS on a lower level compared to the reference scenario in 4.1.1. Over the 40-year modeling time horizon, total investment cost along the CCTS value chain total 145 billion euros. Capture investment occurs in two waves, the first in industry in 2025 and the second in the power sector in 2040 (see Figs. 6 and 9). This is a delay of 5 years compared to the reference scenario (see Figs. 3 and 4). Given a total quantity of avoided emissions of 7.5 $GtCO_2$, this breaks down to investment costs of €19.4 per tCO_2 avoided, an increase to the reference scenario of 22 %. Total variable costs sum up to 266 billion euros, or €35.4 per tCO_2 avoided. With only a slightly higher participation of the power sector, this increase in the average variable costs of CO_2 abatement compared to the reference scenario is explained by longer transport distances and more expensive offshore storage (Figs. 7 and 8).

We also note a lower participation of the cement sector in CCTS whereas capture in the iron and steel sector remains at the same level. This is explained by higher capture costs in the cement industry (see Table 2), but also by the geographical distribution of industrial facilities: while emitters in the iron and steel sector are generally located close to the coast, cement kilns are often located close to inland mining facilities. Thus, a possible strategy could be to form regional clusters that could more easily benefit from economies of scale in transport (Fig. 9).

Results of an offshore only scenario for Germany on a much higher resolution (distance between nodes only 50 km) show a greater drop in CCTS deployment compared to the results presented in this paper [41]. The primary reason for that is the distance of 200 km in between nodes which strongly overestimates economics of scale in transportation since many emitters are grouped and also often set closer to storage sites than in reality. Yet the distance, and therefore the total number of nodes for a modeling region, is limited by computational runtime which increases exponentially with the number of nodes. The scenarios in this paper, which use 460 nodes, require a runtime between 48 and 72 h on a machine with 8 cores and 30 GB RAM[2].

4.2 Certificate Price Increasing to 50 €/tCO2 in 2050

4.2.1 On- and Offshore Storage

Earlier results of the CCTS-Mod focusing only on Germany show that an increase in the CO_2 certificate price to €50 per tCO_2 leads to an application in industry only [41]. Those findings are confirmed by the CCTS-Mod on the European level as well. The lower costs of capture again lead to investments in the steel industry first, followed by the cement industry 5 years later (see Fig. 10 in the Appendix).

[2] 2x Intel Xeon X5355 2.66 GHz Quad-Core, 8 MB Cache

Table 6 Overview of scenario results

Scenario	Pipeline network km	Stored emissions ($GtCO_2$)	Origin from industry (%)	Storage left in 2050 ($GtCO_2$)	CCTS inv. costs (€bn)	CCTS var. costs (€bn)
Ref75	20,400	15.8	63	78.2	240	515
Off75	9,800	7.5	65	42.5	145	266
On50	6,600	5.6	100	88.4	81.4	134
Off50	4,300	2.1	100	47.9	40	58
On100	23,600	24.7	53	69.3	380	929
Off100	37,400	19	57	31	359	796

Source: Own calculation

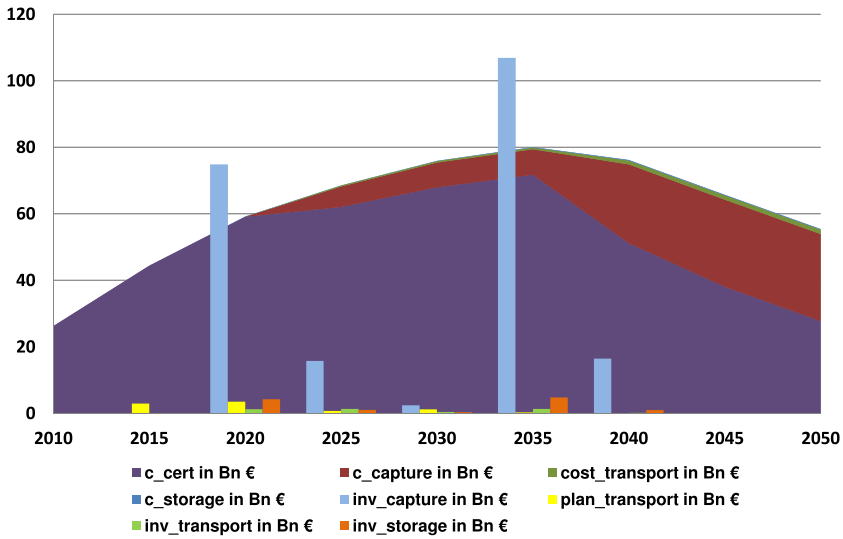


Fig. 4 Infrastructure investment and variable costs in billion €, Ref75

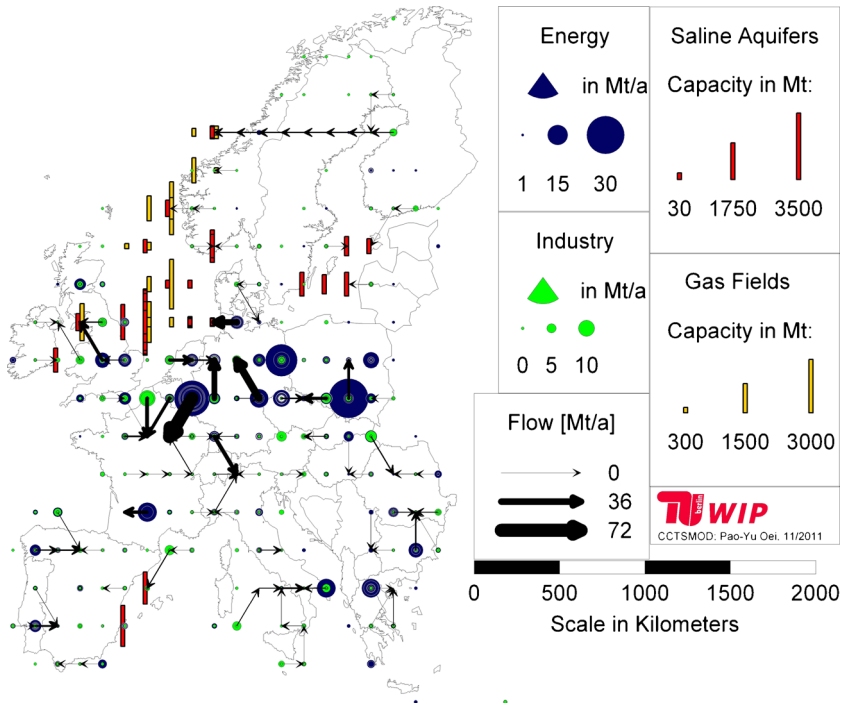


Fig. 5 CCTS infrastructure in 2050, Ref75

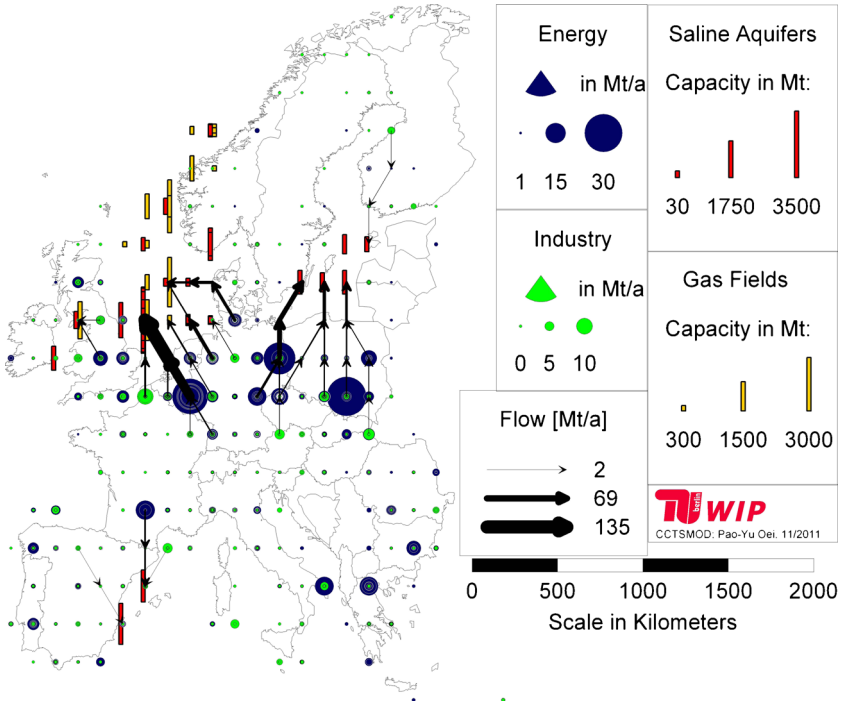


Fig. 6 CCTS infrastructure in 2050, Off75

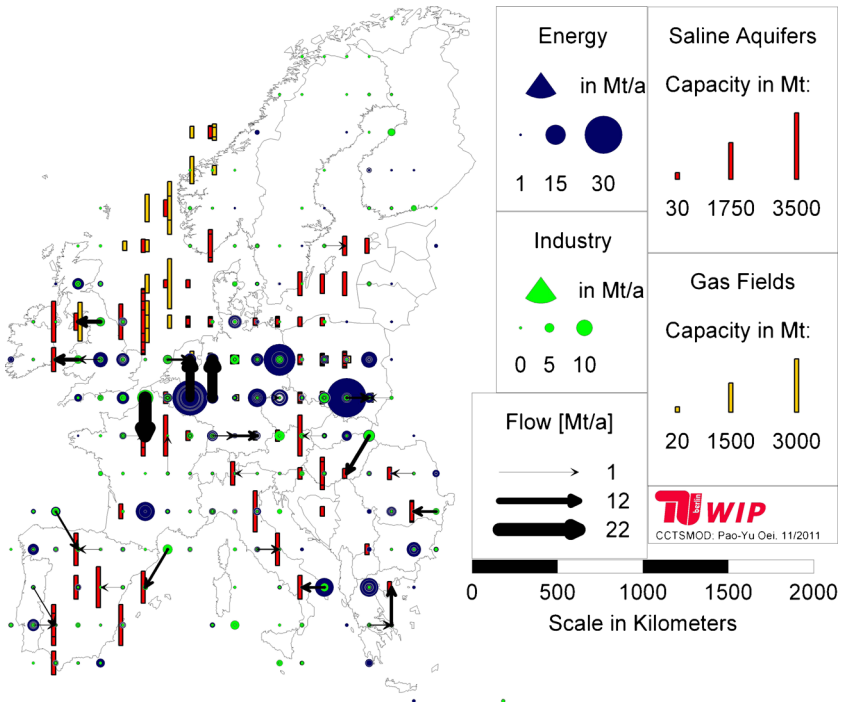


Fig. 7 CCTS infrastructure in 2050, On50

The CCTS technology primarily remains an abatement option for large industry clusters with a regional proximity to storage sites in Northern Europe. This excludes small and mid-scale facilities in the European hinterland. However, with a total storage of 5.6 $GtCO_2$ over the next 40 years, this scenario shows the potential for CCTS in the iron and steel and cement sector even at a low CO_2 certificate price. Investment cost along the CCTS

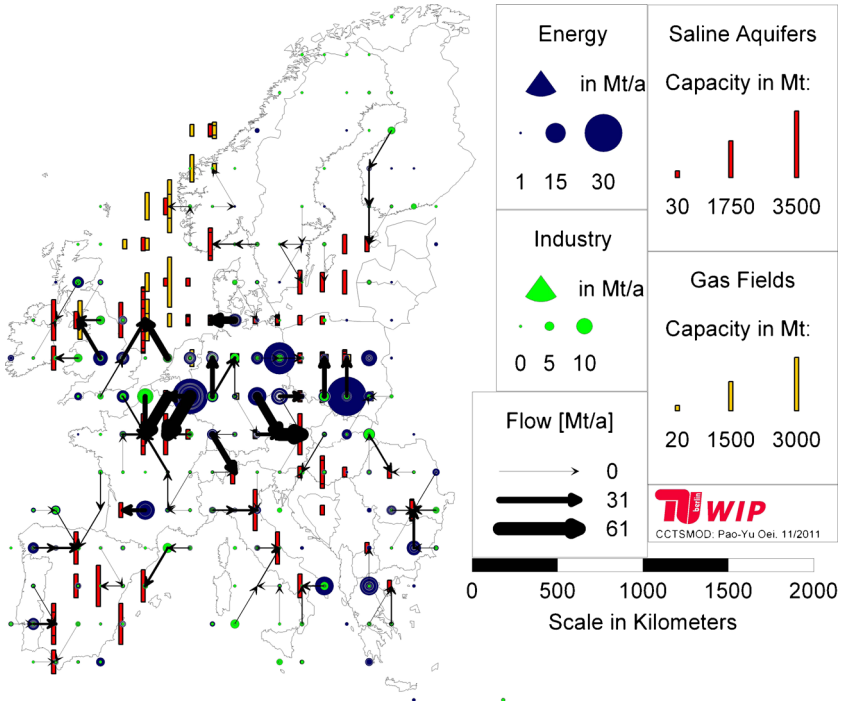


Fig. 8 CCTS infrastructure in 2050, On100

value chain totals 81.4 billion euros. This leads to average investment costs of €14.6 per tCO_2 avoided. Total variable costs sum up to 134 billion euros, or €24 per tCO_2 avoided.

4.2.2 Offshore Storage Only

In the case of offshore storage only, the total storage is reduced to 2.1 $GtCO_2$ over the next 40 years at total investment cost of 40 billion euros. This leads to average investment costs of €18.5 per tCO_2 avoided. Several industrial facilities are located far from any offshore site and thus do not invest in CCTS. They are relatively scattered and cannot form large enough emission clusters to benefit from economies of scale with transporting the CO_2 over longer distances. The total variable costs sum up to 58 billion euros, or €26.4 per tCO_2 avoided. Average variable costs are much lower in case of the certificate price remaining below €50 as the high cost power sector is not investing in the CCTS technology.

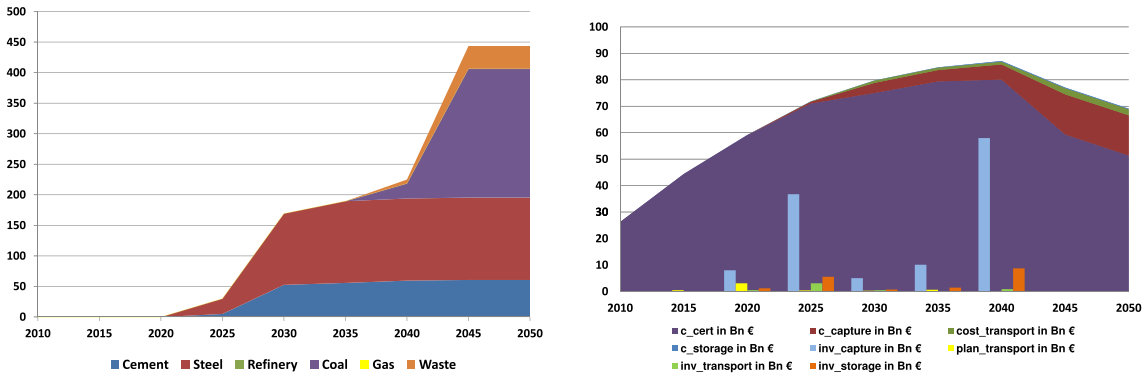


Fig. 9 Storage by sector in million tCO_2 and infrastructure investment and variable costs in billion €, Off75

This scenario highlights the importance of available onshore sinks, especially for the promotion of the CCTS technology at moderate CO_2 prices. However, the current debate on onshore storage in several European countries (e.g., the Netherlands and Germany) indicates that this storage option could be ruled out by regulation (Fig. 11 and 12).

4.3 Certificate Price Increasing to 100 €/tCO2 in 2050

4.3.1 On- and Offshore Storage

This scenario results in a total storage of 24.7 $GtCO_2$ over the next 40 years, a significant increase compared to the reference scenario. The same is true for the investment costs along the CCTS value chain, which increases to 380 billion euros. This leads to average investment costs of €15.4 per tCO_2 avoided. Total variable costs increase to 929 billion euros, or €38 per tCO_2 avoided. This can be explained primarily by the higher participation of the power sector in CCTS.

4.3.2 Offshore Storage Only

This scenario results in a total storage of 19 $GtCO_2$ over the next 40 years. The investment cost along the CCTS value chain add up to 359 billion euros or an average of €18.7 per tCO_2 avoided. The total variable costs are 796 billion euros or €41.5 per tCO_2. The cost increase is based on longer transport distances and the greater participation of the power sector.

5 Conclusion

The role of carbon capture, transport, and storage (CCTS) in future decarbonization portfolios is highly uncertain. Part of this uncertainty is due to a lack of objective information and non-partisan economic analysis. To improve the situation, we develop a model suggesting optimal strategies for deploying a carbon capture, transport, and storage infrastructure. The model integrates a high level of technical detail, while focussing on a simple decision rule on whether "to capture or not to capture": emitters can pay a given CO_2 price, or else engage into CCTS to abate their CO_2; the model will minimize the costs of both, purchase of CO_2 certificates, and CCTS-infrastructure.

With respect to the existing literature, we include new features into the model, such as the explicit recognition of planning costs, as well as the option to combine CCTS in the industry and the electricity sectors. The model suggests that under certain extreme assumptions, CCTS may contribute to the decarbonization of Europe's industry and energy sectors. However, only if the CO_2 certificate price rises to €75 by 2050 and sufficient CO_2 storage capacity is available both on- and offshore, will CCTS have the potential to play a role in future energy technologies. Table 6 provides a summary of the scenario results, in terms of the required pipeline network, total stored emissions, the share of emissions that originate from industrial sources, remaining storage potential, as well as total costs for CCTS (fixed and variable).

Our results indicate that given a moderate increase in the CO_2 certificate price, deployment will be limited only to industrial applications in the iron and steel as well as cement sectors. The infrastructure will remain regional without Europe-wide integration. However, European cooperation could still be of benefit in areas where emission sources and sinks are divided by national borders and for offshore storage solutions.

In all scenarios, industry plays an important role as a first mover to induce deployment. A decrease of available storage capacity or a lower increase in future CO_2 certificate prices could significantly reduce the role of CCTS as a CO_2 mitigation technology, and especially its role in the decarbonization of the electricity sector. We also observe an initial decline in per unit expenditures for CO_2 transport in scenarios with broad CCTS utilization due to economies of scale. In later periods, this effects is, however, partly offset by increasing transport distances due to the development of more distant storage resources, once the close and cheap ones are exhausted. In this context, the storage capacity left at the end of the modeling horizon in 2050 might also be misleading, at first sight. In a post-2050 horizon, cheap storage resources are used up and more distant and costly storage sites will need to be developed. On the other hand, experience gained with developing and operating CO_2 storage sites can also modulate this costs escalation or even lead to overall cost reductions. A quantification of the different effects is up to future research. Another aspect often being neglected is the need for reserving affordable storage options when hoping for negative CO_2 emissions through Biomass–CCTS in some decades. Such and other competing concepts for utilizing underground resources (compressed air storage, natural gas/oil storage, geothermal power, and/or heat recovery) make it difficult to estimate the remaining usable storage potential for CCTS.

Given continued social and political opposition to onshore storage, CO_2 abatement by means of CCTS currently seems only viable with respect to offshore storage. We suggest that policy-makers give first priority to CCTS for coastal areas and small industrial sites where CO_2 transport does not require intensive infrastructure investments to prove the technology's viability, especially in the industry sector. The additional costs of longer pipelines and higher costs for storage development in all offshore scenarios lead to a delay in the CCTS implementation of at least 5 years. However, in reality, this could well be offset by shorter planning processes if the public accepts offshore transport and storage.

Note that our model runs assume a single planner basing its investment decision on full insights into remaining storage capacities, the future CO_2 price development, and actions of all other emitters. The outcomes therefore overestimate the potential for CCTS investment. The key uncertainty of the model is the CO_2 certificate price; its influence on the CCTS-deployment can be seen in the different scenario runs. The variable capturing costs are the second biggest uncertainty of the model and are mainly driven by the electricity price. An increase of these capturing costs would slow down the deployment of CCTS. Transport costs sum up to 10 %, while storage costs lie below 5 % of the overall CCTS-costs in all onshore scenarios. These figures, however, nearly double in the offshore scenarios. Mapping emission sources and sinks to nodes also affects the results, mainly by underestimating the necessary transport infrastructure and overestimating economies of scale (as shown by high-resolution modeling as done for Germany with a distance between nodes of 50 km [41]). Future research should focus on advanced modeling techniques reducing model runtime to enable a European model run with a higher resolution.

Our scenario analysis underlines that the future development of an integrated CCTS infrastructure is highly sensitive to assumptions regarding the future CO_2 certificate prices and the availability of storage resources. If CCTS is to become a cornerstone of a future low carbon industry and power generation sector, policy makers have to commit to clear and reliable targets regarding the future CO_2 prices, or provide alternative long-term investment incentives. Getting the industry sector back into the CCTS debate will help to change the public opinion towards CCTS, when confronted to the lack of alternatives. Based on the persistent experience of canceled and postponed CCTS demonstration projects and reluctant institutional and private investors, the authors doubt that CCTS will become the integrated pan-European industry once envisioned by EU-level policy makers.

Appendix

Table 7 Definition of indices, parameters, and variables of CCTS-Mod

Indices	Description
a, b	Model period
P	Individual CO_2 producer
S	Individual CO_2 storage site
i, j	Node
d	Pipeline diameter [m]
Parameters	
r	Rate of interest [%]
$year_a$	Starting year of a model period
$start$	Starting year of the model
c_ccs_{Pa}	Variable costs of CO_2 capture for producer P [€/t CO_2 per year]
$c_inv_x_P$	Investment costs of CO_2 capture for producer P [€/t CO_2 per year]
$capt_rate$	Capture rate for CO_2 capture of 90%
$CO2_{Pa}$	Total annual quantity of CO_2 produced by producer P [t CO_2]
$cert_a$	CO_2 certificate price [€/ t CO_2]
c_f	CO_2 flow costs [€/t CO_2 per year]
$c_inv_f_d$	Pipeline investment costs [€/km · m (diameter)]
c_plan	Pipeline planning and development costs [€/km]
cap_d_d	Capacity of a pipeline with diameter d [t CO_2/a]
max_pipe	Maximum number of pipelines built along planned route
$c_inv_y_{Sa}$	Investment costs for storage in sink S [€/t CO_2 per year]
cap_stor	Storage capacity of sink S [t CO_2]
$match_P_{Pj}$	Mapping of producer P to node j {0,1}
$match_S_{Sj}$	Mapping of sink S to node j {0,1}
E_{ij}	Distance matrix of possible connections between nodes i and j
Variables	
x_{Pa}	Quantity of CO_2 captured by producer P [t CO_2/a]
inv_x_{Pa}	Investment in additional CO_2 capture capacity [t CO_2/a]
z_{Pa}	Quantity of CO_2 emitted into the atmosphere [t CO_2/a]
f_{ija}	CO_2 flow from node i to j [t CO_2/a]
inv_f_{ijda}	Investment in additional pipeline capacity with diameter d
$plan_{ija}$	Pipeline planning and development between nodes i and j
y_{Sa}	Quantity of CO_2 stored per year in sink S [t CO_2/a]
inv_y_{Sa}	Investment in additional injection capacity of sink S [t CO_2/a]

Table 8 Estimated CO_2 storage potential

Country	Saline aquifer (GT CO_2)	Depleted gasfield (GT CO_2)	Offshore aquifer (GT CO_2)	Offshore gasfield (GT CO_2)	Total (GT CO_2)
Austria	2.30				2.30
Belgium	0.30				0.30
Bulgaria	1.70				1.70
Bosnia and Herzegovina	0.20				0.20
Czech Republic	0.70				0.70
Germany	3.80	1.60	1.20		6.60
Denmark			2.50		2.50
Spain	11.00		3.50		14.50
France	5.70				5.70
Greece	0.30				0.30
Croatia	2.80				2.80
Hungary	0.20				0.20
Ireland	2.00		1.30		3.30
Italy	5.50				5.50
Latvia			1.30		1.30
Macedonia	0.30				0.30
The Netherlands		0.70		0.50	1.20
Norway			1.90	11.90	13.80
Poland	3.70	0.70	3.50		7.90
Romania	0.40				0.40
The United Kingdom			14.40	7.80	22.20
Total	40.90	3.00	29.60	20.20	93.70

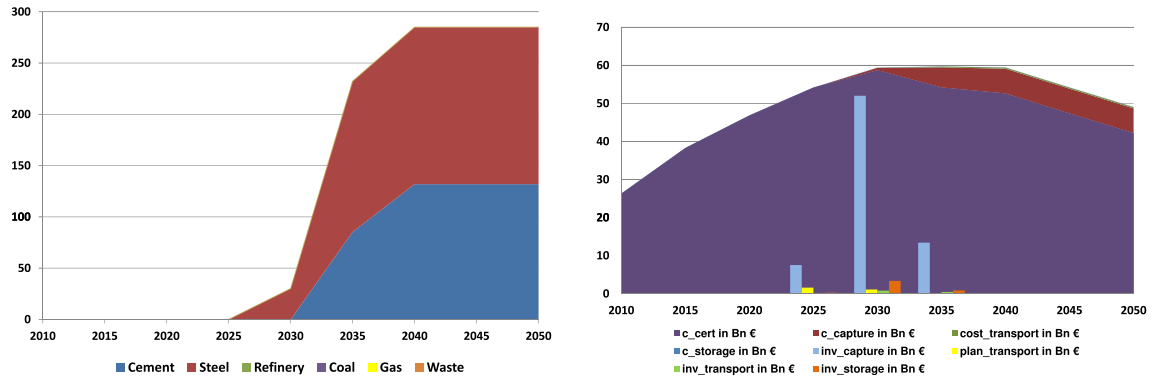


Fig. 10 Storage by sector in million tCO_2 and infrastructure investment and variable costs in billion €, On50

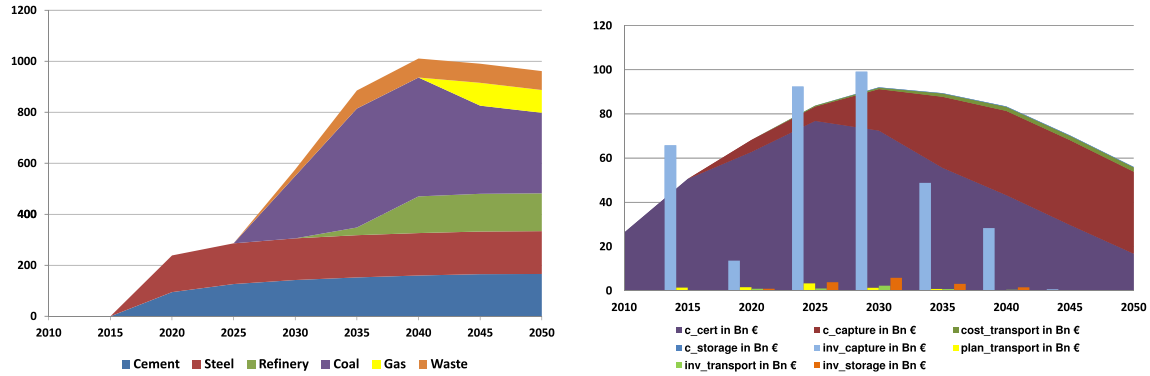


Fig. 11 Storage by sector in million tCO_2 and infrastructure investment and variable costs in billion €, On100

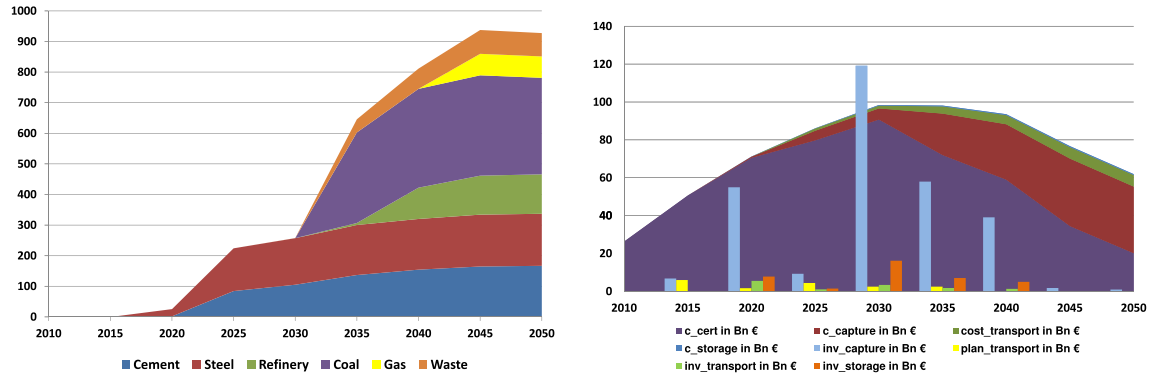


Fig. 12 Storage by sector in million tCO_2 and infrastructure investment and variable costs in billion €, Off100

References

1. Ainger, D., Argent, S., Haszeldine, S. (2010). October 2010. Europe-wide CO_2 infrastructures feasibility study. *Berlin Forum on Sustainable Fossil Fuels*, 18–19.
2. ARI, M.-C. (2010). Optimization of CO_2 storage in CO_2 enhanced oil recovery projects. Tech. rep., Department of energy & climate change (DECC), office of cabon capture and storage.
3. Bentham, M. (2006). An assessment of carbon sequestration potential in the UK--Southern North Sea case study. Working Paper 85, Tyndall Centre for Climate Change Research.
4. Bentham, M., Kirk, K., Wiliams, J. (2008). Basin-by-basin analysis of CO_2 storage potential of all-island Ireland. Sustainable energy and geophysical surveys programm. Commissioned report CR/08/036, British geological survey.
5. Brook, M., Shaw, K., Vincent, C., Holloway, S. (2009). Gestco case study 2a-1: storage potential of the bunter sandstone in the UK sector of the southern North Sea and the adjacent onshore area of Eastern England. Tech. rep., sustainable energy and geophysical surveys programm. commissioned report CR/03/154.
6. Davidson, C., Dahowski, R., Dooley, J. (2011). A quantitative comparison of the cost of employing EOR-Coupled CCS supplemented with secondary DSF storage for two large CO_2 point sources. *Energy Procedia*, *4*(January), 2361–2368.
7. Dooley, J.J., Dahowski, R.T., Davidson, C.L., Wise, M.A., Gupta, N., Kim, S.H. (2006). In *Carbon dioxide capture and geological storage—a core element of a global energy technology strategy to address climate change*. USA: College Park.
8. EC (2012). 23 Innovative renewable energy demonstration projects receive € 1.2 billion EU funding. European commission, climate action, newsroom, brussels, 18. December.
9. EEA. (June 2011). In *The European Pollutant Release and Transfer Register*. Copenhagen, Denmark: European Environmental Agency.
10. Finkenrath, M. (2011). In *Cost and performance of cabon dioxide capture from power generation. Tech. rep., International energy agency (IEA/OECD)*. Paris: France.
11. Fritze, K. (2009). Modeling CO_2 storage pipeline routes in the United States. Tech. rep., Nicholas School of the Environment and Earth Sciences. http://hdl.handle.net/10161/985.
12. GeoCapacity, E. (2009). In *Assessing european capacity for geological storage of carbon dioxide. Tech. rep.,EU GeoCapacity—Geological Survey of Denmark and Greenland*.
13. Gerling, J.P. (2010). In *CO_2 Storage—German and International Perspective. Berlin seminar on energy and climate policy, Deutsches Institut fr Wirtschaftsforschung* Vol. 3. Berlin: Juni 2010.
14. Greenpeace (2011). Potentielle CO_2-Endlager in Deutschland in Salzwasser fhrenden Tiefengestein (in German). Auswertung der BGR-Daten durch Greenpeace.
15. Hazeldine, S. (March 2009). Carbon capture and storage: UK's fourth energy pillar, or broken bridge? Working Paper SCCS 2009-03. Scottish Centre for Carbon Storage.

16. Heddle, G., Herzog, H., Klett, M. (2003). The Economics of CO_2 Storage. MIT LFEE 2003-003 R. Massachusetts Institute of Technology.
17. Herold, J., Ruester, S., von Hirschhausen, C. (2010a). Vertical integration and market structure along the extended value added chain including carbon capture, transport and sequestration (CCTS). Project report for the european commission. *Project No*, 213744. SECURE Work package 5.3.2.
18. Herold, J., von Hirschhausen, C., Ruester, S. (2010b). In *Carbon capture, transport and storage in Europe: a problematic energy bridge to nowhere? CEPS Working Document 341*. Belgium: Brussels.
19. Hirschhausen, C.v., Herold, J., Oei, P.-Y. (2012). How a low carbon innovation can fail—tales from a lost decade for carbon capture, transport, and sequestration (CCTS). *Economics of Energy & Environmental Policy*, *2*, 115–123.
20. Ho, M.T., Allinson, G.W., Wiley, D.E. (2010). Comparison of MEA capture cost for low CO_2 emissions sources in Australia. http://www.sciencedirect.com/science/article/B83WP-50G5H9G-3/2/c46881a78b698c9ca910a8dafdb39877.
21. IEA. (2005). In *Building the cost curves for CO_2 storage: European sector. Tech. rep., international energy agency*. France: Paris.
22. IEA. (2009). In *CO_2 Capture and storage—a key carbon abatement option. Tech. rep., international energy agency*. France: Paris.
23. IEA. (2012). In *Energy Technology Perspectives 2012. Pathways to a clean energy system. Tech. rep., international energy agency*. France: Paris.
24. IEA, & UNIDO. (2011). In *Technology roadmap carbon capture and storage in industial applications. Tech. rep., international energy agency. united nations industrial development organization*. France: Paris.
25. IEAGHG, Z.EP. (2011). In *The costs of CO_2 storage—post demonstration CCS in the EU. Tech. rep., IEA greenhouse gas R & D programme (IEAGHG) and european technology platform for zero emission power plants (ZEP)*. Belgium: Brussels.
26. IETA (2012). Briefing on the EU's emissions trading scheme. Tech. rep.,International Emissions Trading Association.
27. IPCC (2005). IPCC special report on carbon dioxide capture and storage. Tech. rep., intergovernmental panel on climate change (IPCC), Internet:Accessed: 31 July 2009.
28. Kazmierczak, T., Brandsma, R., Neele, F., Hendriks, C. (2008). In *Algorithm to create a CCS low-cost pipeline network. Energy Procedia 1 (1), 1617–1623, greenhouse gas control technologies 9, proceedings of the 9th international conference on greenhouse gas control technologies (GHGT-9)*, (pp. 16–20). Washington DC, USA: November 2008.
29. Kemp, A., & Kasim, S. (2012). The Economics of CO_2-EOR Cluster Developments in the UK Central North Sea/Outer Moray Firth. *North Sea Study Occational Paper*, *123*, 1–64.
30. Klokk, O., Schreiner, P., Pagès-Bernaus, A., Tomasgard, A. (2010). Optimizing a CO_2 value chain for the Norwegian continental shelf. *Energy Policy*, *38*, 6604–6614.
31. Öko Institut. (2012). In *Potenziale und Chancen der Technologie zur CO_2- Abtrennung und Ablagerung (CCS) fr industrielle Prozessemissionen [in German]. Study for the world wildlife. Tech. rep.*
32. Kobos, P.H., Malczynski, L.A., Borns, D.J., McPherson, B.J. (2007). The 'String of Pearls': The Integrated Assessment Cost and Source-Sink Model. The 6th Annual Carbon Capture & Sequestration Conference, Pittsburgh, PA, May 7–10, 2007.
33. Kuby, M.J., Bielicki, J.M., Middleton, R.S. (2011). Optimal spatial deployment of carbon dioxide capture and storage given a price on carbon dioxide.
34. Kuramochi, T., Ramírez, A., Turkenburg, W., Faaij, A. (2012). Comparative assessment of CO_2 capture technologies for carbon-Intensive industrial processes. *Progress in Energy and Combustion Science*, *38*(1), 87–112.
35. McPherson, B., Allis, R., Biediger, B., Brown, J., Cappa, G. (2009). In *Southwest regional partnership on carbon sequestration. Revised semiannual report. New Mexico institute of mining and technology, Socorro*. USA: New Mexico.
36. Mendelevitch, R. (2013). The Role of CO_2-EOR for the Development of a CCTS Infrastructure in the North Sea Region: A Techno-Economic Model and Application. Discussion Papers of DIW Berlin. *German Institute for Economic Research*, *1308*, 1–51.
37. Middleton, R., Herzog, H., Keating, G., Kuby, M., Liao, X. (2007). Optimization for geologic carbon sequestration and carbon credit pricing. Sixth annual conference on carbon capture & sequestration.
38. Middleton, R.S., & Bielicki, J.M. (2009). A scalable infrastructure model for carbon capture and storage: SimCCS. *Energy Policy*, *37*(3), 1052–1060.
39. MIT (2012). Carbon capture and sequestration technologies @ MIT. Carbon apture and sequestration project database. Tech. rep., Massachusetts Institute of Technology. URL http://sequestration.mit.edu/tools/projects/index_capture.htmlAccessed: 7thOctober2013.
40. Neele, F., Hendriks, C., Brandsma, R. (Feb. 2009). Geocapacity: economic feasibility of CCS in networked systems. *Energy Procedia*, *1*(1), 4217–4224. URLhttp://www.sciencedirect.com/science/article/pii/S1876610209008753.
41. Oei, P.-Y., Herold, J., Tissen, A. (2011). CO_2 Speicherung in Deutschland: Eine Brckentechnologie als Klimaloesung? [in German]. *Zeitschrift fr Energiewirtschaft*, *35*(4), 263–273.
42. Platts (January 2011). Power system database Europe. The McGraw-Hill companies.
43. Radoslaw, T., Barbara, U.-M., Adam, W. (2009). CO_2 storage capacity of deep aquifers and hydrocarbon fields in Poland—EU GeoCapacity project results. *Energy Procedia*, *1*(1), 2671–2677.
44. RECCS. (2010). In *RECCS plus: comparison of renewable energy technologies with carbon dioxide capture and storage. Tech. rep., federal ministry for the environment, Nature Conservation and Nuclear Safety (BMU)*. Germany: Berlin.
45. Rubin, E.S., Yeh, M.A., Berkenpas, M., Davison, J. (2007). Use of experience curves to estimate the future cost of power plants with CO_2 capture. *International Journal of Greenhouse Gas Control*, *1*(2), 188–197.
46. Tavoni, M., & van der Zwaan, B. (2011). Nuclear Versus Coal plus CCS: a comparison of two competitive base-Load climate control options. *Environmental Modeling and Assessment*, *16*, 431–440.
47. Tzimas, E. (2009). In *The cost of carbon capture and storage demonstration projects in Europe. Tech. rep., JRC Scientific and Technical Reports, European Commission*.
48. VGB (2011). Geplante Neubauprojekte in der EU [in German]. Tech. rep. *VGB PowerTech e*, *V*.
49. WorleyParsons, S.chlumberger. (2011). In *Economic assessment of carbon capture and storage technologies: 2011 update. Tech. rep., The Global CCS Institute*. Australia: Camberra.
50. ZEP. (2011a). In *The costs of CO_2 Capture. Post Demonstration CCS in the EU. Tech. rep., European technology platform for zero emissions fossil fuel power plants. Zero Emissions Platform*. Belgium: Brussels.
51. ZEP. (2011b). In *The costs of CO_2 transport. Post demonstration CCS in the EU. Tech. rep., European technology platform for zero emissions fossil fuel power plants. zero emissions platform*. Belgium: Brussels.